

ECON 6018

TUTORIAL 4

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- Lesson Tutorial
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Footnote: The practice questions are adapted from Essential Mathematics for Economic Analysis (5th Edition) by Knut Sydsaeter, Peter Hammond, Arne Strøm, and Andres Corgial. All credit for the original questions belongs to the authors. The solutions and explanations provided here are my own work.

Focus :

1. Partial Derivatives
2. Introduction to Implicit
3. Practices to improve speed

Chapter 11 : Partial Derivatives

If $z = f(x, y)$

$\frac{\partial z}{\partial x}$ derivative of f wrt x when y is held constant

$\frac{\partial z}{\partial y}$ derivative of f wrt y when x is held constant

Let us walk through an example together!

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

(a) $z = 2x + 3y$ (b) $z = x^2 + y^3$ (c) $z = x^3 y^4$ (d) $z = (x + y)^2$

Ans:

$$(a) \frac{\partial z}{\partial x} = 2 \quad \frac{\partial z}{\partial y} = 3$$

$$(b) \frac{\partial z}{\partial x} = 2x \quad \frac{\partial z}{\partial y} = 3y^2$$

$$(c) \frac{\partial z}{\partial x} = 3x^2y^4 \quad \frac{\partial z}{\partial y} = 4x^3y^3$$

Questions:

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

(a) $z = x^2 + 3y^2$ (b) $z = xy$ (c) $z = 5x^4y^2 - 2xy^5$ (d) $z = e^{x+y}$

(e) $z = e^{xy}$ (f) $z = e^{\frac{x}{y}}$ (g) $z = \ln(x+y)$ (h) $z = \ln(xy)$

Ans:

2. (a) $\frac{\partial z}{\partial x} = 2x$

$$\frac{\partial z}{\partial y} = 6y$$

(b) $\frac{\partial z}{\partial x} = y$

$$\frac{\partial z}{\partial y} = x$$

(c) $\frac{\partial z}{\partial x} = 20x^3y^2 - 2y^5$

$$\frac{\partial z}{\partial y} = 10x^4y - 10xy^4$$

(d) $\frac{\partial z}{\partial x} = e^{x+y}$

$$\frac{\partial z}{\partial y} = e^{x+y}$$

(e) $\frac{\partial z}{\partial x} = ye^{xy}$

$$\frac{\partial z}{\partial y} = xe^{xy}$$

(f) $\frac{\partial z}{\partial x} = \frac{1}{y}(e^x)$

$$= \frac{e^x}{y}$$

$$\frac{\partial z}{\partial y} = (e^x) \cdot \frac{1}{y^2}$$

$$= -\frac{e^x}{y^2}$$

(g) $\frac{\partial z}{\partial x} = \frac{1}{x+y}$

$$\frac{\partial z}{\partial y} = \frac{1}{x+y}$$

2. Find $f'_1(x, y)$, $f'_2(x, y)$, $f''_{12}(x, y)$

(a) $f(x, y) = x^7 - y^7$ (b) $f(x, y) = x^5 \ln y$ (c) $f(x, y) = (x^2 - 2y^2)^5$

$$3.(a) f'_1(x, y) = 7x^6$$

$$f'_2(x, y) = -7y^6$$

$$f''_{12} = 0$$

$$(b) f'_1(x, y) = (\ln y)(5x^4) \quad f'_2(x, y) = x^5 \left(\frac{1}{y}\right) \quad f''_{12}(x, y) = \frac{5x^4}{y}$$
$$= 5 \ln y x^4 \quad = \frac{x^5}{y}$$

$$(c) f'_1(x, y) = 5(x^2 - 2y^2)^4 \cdot 2x$$

$$= 10x(x^2 - 2y^2)^4$$

$$f'_2(x, y) = 5(x^2 - 2y^2)^4 \cdot 4y$$

$$= 20y(x^2 - 2y^2)^4$$

$$f''_{12}(x, y) = 10x \cdot 4(x^2 - 2y^2)^3 \cdot -4y$$

$$= -160xy(x^2 - 2y^2)^3$$

3. Find first and second-order partial derivatives

(a) $z = 3x + 4y$ (b) $z = x^3y^2$ (c) $z = x^5 - 3x^2y + y^6$

(d) $z = \frac{x}{y}$ (e) $z = \frac{(x-y)}{(x+y)}$ (f) $z = \sqrt{x^2 + y^2}$

$$4.(a) \frac{\partial z}{\partial x} = 3$$

$$\frac{\partial z}{\partial y} = 4$$

$$f''_{12}(x, y) = 0$$

$$(d) \frac{\partial z}{\partial x} = (1) \left(\frac{1}{y} \right)$$

$$= \frac{1}{y}$$

$$\frac{\partial z}{\partial y} = x \cdot -\frac{1}{y^2}$$

$$= -\frac{x}{y^2}$$

$$f''_{12}(x, y) = -\frac{1}{y^2}$$

$$(b) \frac{\partial z}{\partial x} = 3x^2 y^2$$

$$\frac{\partial z}{\partial y} = 2x^3 y$$

$$f''_{12}(x, y) = 6x^2 y$$

$$(e) \frac{\partial z}{\partial x} = 1 \cdot \left(\frac{1}{x+y} \right) + (x-y)(x+y)^{-2} \cdot -$$

$$= \frac{1}{x+y} - \frac{x-y}{(x+y)^2}$$

$$= \frac{\cancel{x+y} - \cancel{x+y}}{(x+y)^2}$$

$$= \frac{2y}{(x+y)^2}$$

$$(c) \frac{\partial z}{\partial x} = 5x^4 - 6xy$$

$$\frac{\partial z}{\partial y} = -3x^2 + 6y^5$$

$$f''_{12}(x, y) = -6x$$

$$\frac{\partial z}{\partial y} = -1 \cdot \left(\frac{1}{x+y} \right) + (x-y)(x+y)^{-2} \cdot -1$$

$$= \frac{-\cancel{x-y} - \cancel{x+y}}{(x+y)^2} = -\frac{2x}{(x+y)^2}$$

$$f''_{12}(x, y) = 2 \cdot \left(\frac{1}{(x+y)^2} \right) + (2y)(x+y)^{-3} \cdot -2$$

$$= \frac{2}{(x+y)^2} - \frac{4y}{(x+y)^3}$$

$$= \frac{2x+2y-4y}{(x+y)^3}$$

$$= \frac{2x-2y}{(x+y)^3}$$

Application of partial derivatives

- rmb our chain rule?

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

- let's expand this

$$z = f(x, y)$$

$$x = g(t)$$

$$y = h(t)$$

$$z = f(g(t), h(t))$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Let's go through an example!

$$F(x, y) = x + y^2$$

$$x = t^2$$

$$y = t^3$$

Question

$$F(x, y) = x^p y^q$$

$$x = at$$

$$y = bt$$

Introduction to Implicit (Chapter 7)

$$- y = x^2$$

$$\frac{dy}{dx} = 2x$$

notice :

$$\frac{d}{dx} y = \frac{d}{dx} x^2$$

$$\downarrow$$
$$\frac{dy}{dx}$$

$$\text{SO: } \frac{d}{dx} y = \frac{dy}{dx}$$

what is y^2 wrt. x

$$\frac{d}{dx} y^2 = \frac{dy}{dx} \cdot 2y$$

Example:

$$x^2 y = 1$$

$$\text{Qns: } x - y + 3xy = 2$$